

Data Analytics Lab (CS689)

Lab 6

The objective of this assignment is to understand and implement different regression models using Python and evaluate their performance.

1. Import the dataset.
2. Implement **Simple Linear Regression**.
3. Implement **Multiple Linear Regression**.
4. Implement **Quadratic (Polynomial) Regression**.
5. Evaluate models using **R² and SSE**.
6. Study the effect of **influential points (outliers)** on regression.

Dataset Description

In this assignment, a **synthetic House Price dataset** is to be used. The dataset simulates factors that typically influence housing prices.

Each observation represents a house with the following attributes:

Variable	Description	Range
Size	Total area of the house in square feet	500 – 3000
Bedrooms	Number of bedrooms in the house	1 – 4
Age	Age of the house in years	1 – 30
Price	Market price of the house (target variable)	Generated using regression equation

The price of the house is generated using the following relationship:

$$Price = 50000 + 120(Size) + 15000(Bedrooms) - 800(Age) + Noise$$

Where:

- **Size** increases the price.
- **Bedrooms** positively influence the price.
- **Age** negatively affects the price (older houses tend to be cheaper).
- **Noise** represents random variations in the market.

The dataset contains **100 observations**.